CRITERION B SAMPLE

Title:

Tennis Ball Bounce Height Before and After Two Hours of Play: Which Height will be Greater?

Background Information:

Validity of the Research Question: The purpose of this experiment is to determine which type of tennis ball will bounce higher when dropped under the same conditions: a new, unused ball or the same ball after two hours of play. Knowing how much bounce life a tennis ball will have before and after play is a relevant study because this knowledge can help with the performance and consistency of the game. For example, if a player knows that the bounce height of the ball drops by 10% after two hours of play, then they know when to adjust the contact point so the ball is hit in such a way that a successful return can be made ("Can you give me some reasons")
Reasoning behind the hypothesis: A tennis ball has a rubber core filled with pressurized air (Admin). When a tennis ball is hit, it deforms slightly causing some air to be released from the ball. A tennis ball is made from a special rubber compound. After hours of play, the rubber is compressed and stretched repeatedly. The more the rubber is used, the less likely it is to fully snap back into place ("The Science of Tennis Balls"). The hypothesis goal is to see if this science holds true. The experiment will conduct three trials where each type of tennis ball is dropped from a height of 1 meter on the same surface. The data collected will be analyzed, and it will be determined which ball bounces back at a greater height. Research will be conducted to determine why one ball bounces back at a greater height than the other.

Research Question:

Which type of ball, a new unused tennis ball or the same tennis ball after two hours of play, will bounce higher?

Hypothesis:

If a tennis ball is dropped from the same height before and after two hours of play, **then** the unused tennis ball will bounce back higher **because** the rubber will lose elasticity after two hours of use.

Materials:

- Meter stick (1)
- Roll of double sided tape (1)
- Scissors (1)
- Sharpie marker (1)
- Tennis ball (3 of the same)
- Pickle ball (3 of the same)
- Iphone camera is Slow motion (1)

Variables:

Type of Variable	**Variable in the Experiment**	**Method of Measurement/ Method of Control**
Dependent	The bounce height of the tennis ball	This will be measured using a meter stick attached to the wall and an iPhone camera in slow motion to record the motion
Independent	The type of tennis ball (new or used for 2 hours)	The two types of balls considered are a new tennis ball and a tennis ball used for two hours; a new identical ball will be used in each trial
Control	Drop height of the ball (1 m)	During each trial the ball is dropped from a height of 1 m measured using the same meter stick
Control	Surface the ball is dropped on	During each trial the ball is dropped on the same tile floor

Control	Method of recording the ball drop	During each trial the iPhone camera was set to the same setting on slow motion and the same iPhone camera was used

Procedure:

1. A meter stick was placed vertically against a wall 2. A scissors were used to cut six pieces of double sided tape from the roll 3. The six pieces of double sided tape were placed on the back of the meter stick 4. The meter stick was attached to the wall using the tape and a level was used to ensure that the meter stick was attached evenly 5. An iPhone camera was set up to film in slow motion 6. The iPhone camera was started in slow motion at the same time that the new tennis ball was dropped from the top of the meter stick (the 1 m mark) 7. Once the tennis ball stopped bouncing, the camera was stopped 8. The film of the bounce was observed and the data was recorded in Table 1 9. Steps 5 to 8 were repeated two more times using a different identical new tennis ball each time 10. Steps 5 to 9 were repeated using the tennis ball after two hours of use instead of a new tennis ball